Role of Neuroinflammation and Gender Differences in Alzheimer's Disease

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Background

Alzheimer's disease (AD) is a devastating progressive neurodegenerative disease resulting in memory loss, behavioral deficits, and ultimately loss of normal bodily functions. It is estimated that an average of 5.4 million Americans currently have Alzheimer's disease (Alzheimer's Association, 2016). Within the next 30 years, that number is expected to grow to over 13 million people afflicted with AD—there is still no known cure.

The vast majority of AD cases are Late-onset AD, when symptoms display after the age of 65. Late-onset AD is caused by sporadic AD. The rate of progression varies between patients, while most AD patients live on average 8 years from diagnosis of symptoms, individuals may live as long as 20 years with AD.

The anatomical progression of AD occurs in a predictable pattern, with initial amyloid-β and tau aggregation beginning in the medial temporal lobe (entorhinal cortex) and then hippocampus, as well as the prefrontal cortex . During the later stages pathology markers spread throughout the cortex and limbic structures.

Many physiological variables contribute to AD onset/progression, including: amyloid-beta plaques, tau tangles, inflammation, vascular dysfunction (infarcts) and glucose intolerance. Protein dysfunction and aggregation are the hallmark physiological symptoms of AD, represented by amyloid-β plaque formation and neurofibrillary tau tangles.

Neuroinflammation during AD progression exacerbates pathogenesis and leads to disruptions in cell signaling and ultimately neuronal death. However, it has not been established whether early exposure to inflammation may play a significant role in AD onset and progression. It is also unknown whether inflammation affects females differentially than males.

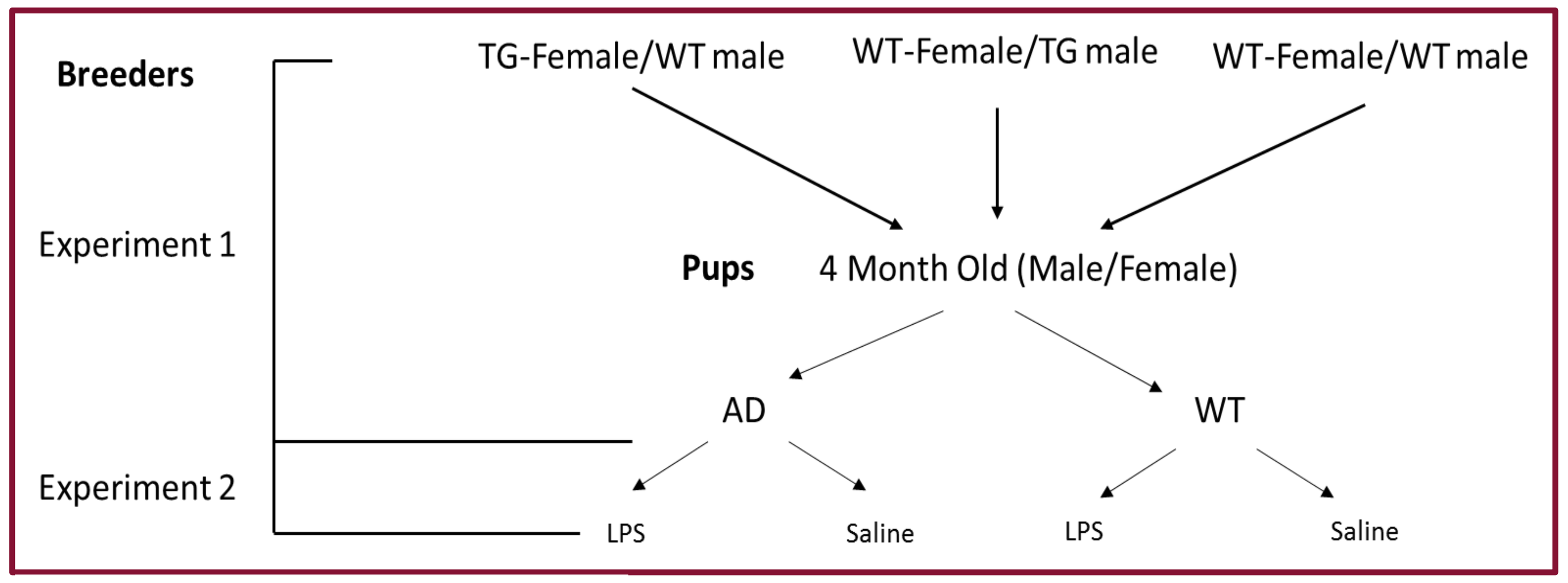
In the United States, women make up over 60% of the AD demographic (Mufson, 2016). Women also show a faster rate of progression as measured by psychometric test batteries and Braak stages (Sperling, 2011). Gender differences, particularly in late-adulthood, may contribute to the chronic inflammatory state found in AD. This may underlie the heightened risk and severity for women later in life, even when longevity is factored in.

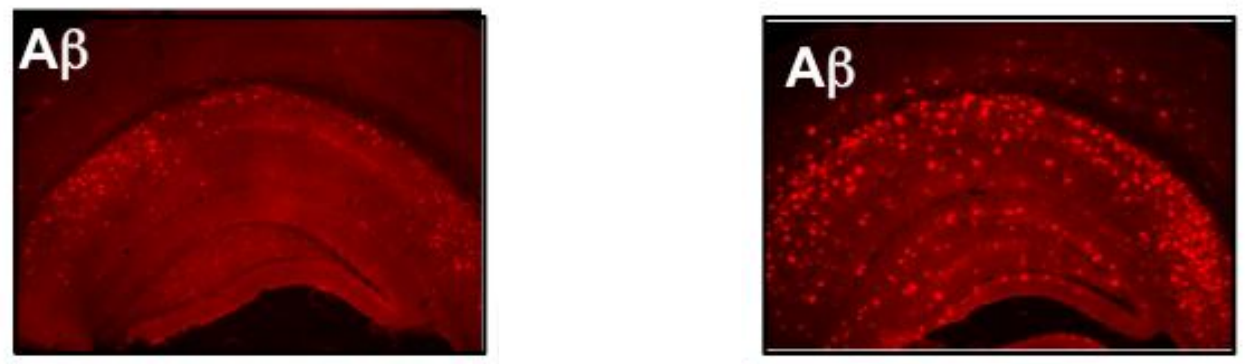
Hypothesis-Early exposure to inflammatory stress worsens AD onset and progression, specifically in females compared to males.

Aim 1- Demonstrate the effect of inflammation timing on AD onset and progression

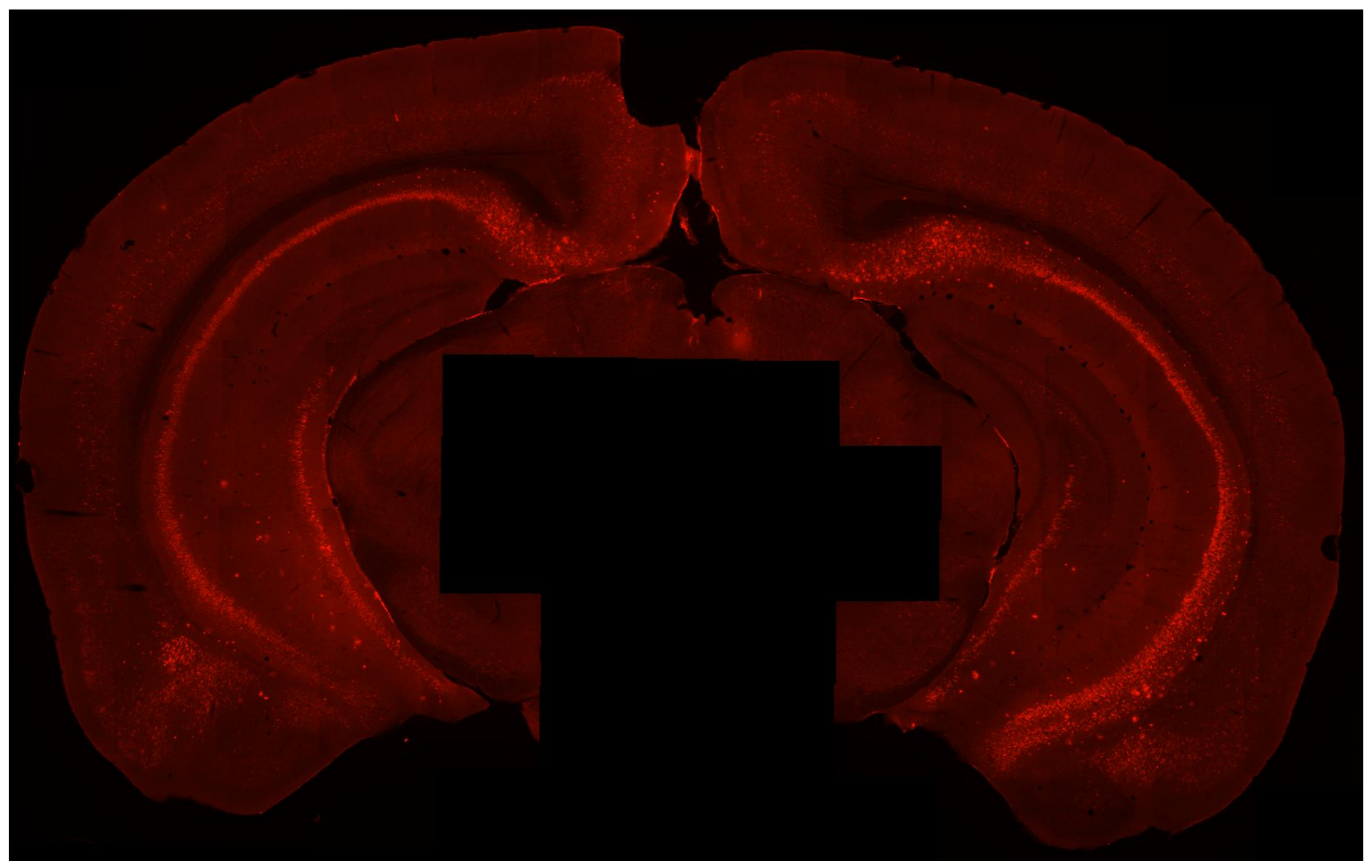
Aim 2-Determine if gender differences are present during AD onset and progression.

Experimental Design

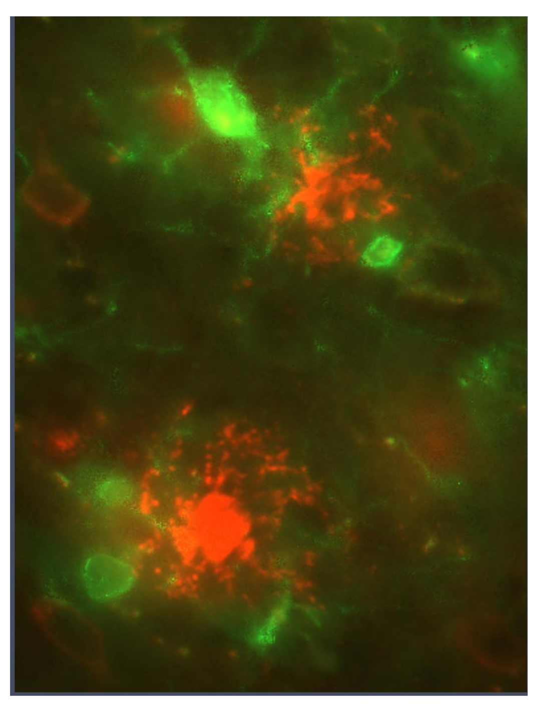




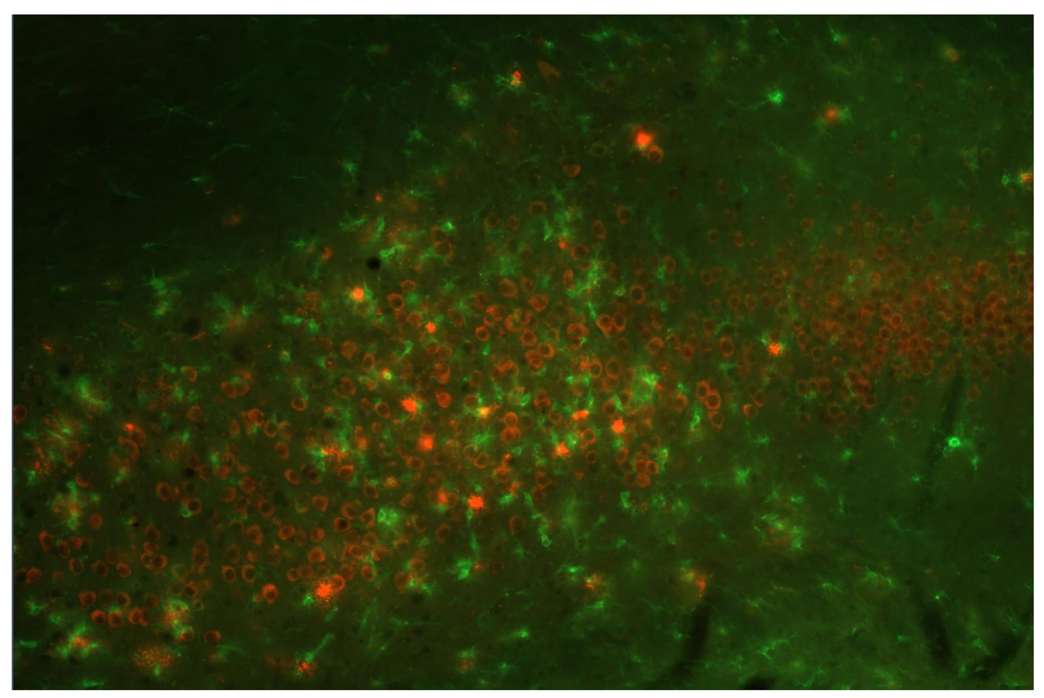
Hippocampal amyloid-beta plaque formation (6E10) at 2 months of age (left) and 6 months of age (right).



Amyloid-beta (6E10) staining of coronal section of the caudal part of the hippocampus and surrounding cortical regions at 4 months of age



Cortical microglia (Iba1) in green surrounding plaques (6E10) in red at 40x objective.



Hippocampal microglia (Iba1) in green and amyloid-beta/plaque (6E10) formation in red at 20x objective.

Material and Methods

The 5 X's Familial Alzheimer's disease (5xFAD) transgenic mouse model was used in this study. This model incorporates 3 human amyloid precursor protein (APP) mutations and 2 human presenilin 1 (PSEN1) mutations that produce amyloid-β deposition and ultimately senile plaque formation. Amyloid-β production influences a pro-inflammatory state—marked by microglia activation.

Histological analysis using 30mm vibratome sections were used to quantify amyloid-beta (6E10) expression in the hippocampus and surrounding cortical regions, as well as microglia (Iba1) count in this area. Unbiased stereology software will be used to quantify the presence of plaques and microglia.

In experiment 1, early exposure to inflammation will be observed based on the phenotype of the mother. TG females display plaque formation and chronic inflammation which exposes the pups prenatally to inflammatory stressors before the onset of AD. Amyloid- β dysfunction does not occur until day 7 due to the presence of the Thy1 promotor. Therefore, pups subjected to inflammation early will be primed for a postnatal inflammatory response before the onset of AD physiology and symptoms. Mice born from wild type mothers are not subjected to chronic inflammation before the onset of AD.

In experiment 2, Lipopolysaccharide (LPS) was injected intraperitonially at 2 months of age to create a chronic inflammatory state. LPS readily crosses the blood brain barrier and causes a long lasting inflammation and activation of microglia.

Perspectives

The goal of this study is to determine if inflammation influences the onset and progression of AD. Symptomatology will be measured and analyzed based on plaque and microglia quantity. Early exposure to inflammation (prenatally) may prime mice for a heightened secondary response (Experiment 1). When possible, preventative healthcare practices (i.e.-NSAID administration) may be a reliable and effective treatment option before the manifestation of clinical symptoms occurs. Exposure to inflammation after the onset of AD (experiment 2) will be analyzed to determine if inflammatory stress exacerbates existing conditions after the onset of AD.

This experiment will also provide context as to whether a gender difference during exposure to inflammation occurs. Early exposure to inflammation in females may increase the onset and timing of disease progression compared to their male counterparts. Underlying mechanisms (i.e.-hormonal differences, environmental differences) will need to be further examined.

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